

Образложение за пријава на магистерски труд

A Reproducible Methodology for Generating Modern Intrusion Detection Datasets

(наслов на магистерскиот труд)

Јован Симоноски

(име и презиме на студентот)

1. Образложение на проблемот и дефинирање на истражувањето

Во овој дел се внесува опис на проблемот, теоретски основи, преглед на досега објавени достигања во областа и дефинирање на значењето на истражувањата спроведени во наведената област од интерес и цели и мотивација за спроведување на истражувањата.

Machine learning has become one of the most promising approaches for Intrusion Detection and Intrusion Prevention Systems (IDS and IPS). The quality of these systems depends almost entirely on the data they are trained on, and this is where the field has a persistent problem.

The most widely used datasets in IDS research, such as KDD'99, NSL-KDD, UNSW-NB15, and the CICIDS family, were generated years or even decades ago, and the attacks they capture reflect the threat landscape of that time. Many of the exploited vulnerabilities are now patched or obsolete, while modern exploitation techniques and recent CVEs, particularly those from the last few years, are often absent. Models trained on this kind of data tend to perform impressively in published benchmarks but struggle against the kinds of attacks defenders face today.

Another important aspect is that the process by which existing datasets were built is rarely documented in sufficient detail in order to be able for others to reproduce it. Decisions about environment and architecture design, telemetry collection, time synchronization, labeling, and quality control are often described only in broad strokes, which makes it difficult for other researchers and institutions to produce comparable datasets of their own.

In addition to the many outdated datasets in the field, very few methodologies for creating new ones exist, and as the threat landscape keeps evolving, the absence of such methodologies means that datasets age faster than they can be replaced.

The theoretical basis of the work is based on combination of network security, intrusion detection, and supervised machine learning. The idea is to link observed traffic (flows, packets and logs) to a label which indicates whether it is benign or attack traffic. This linking depends on a realistic environment, a representative set of attacks, and an accurate labelling process.

Earlier work in this field includes foundational contributions by the Canadian Institute for Cybersecurity and the University of New South Wales on the generation of IDS datasets. This research builds on these efforts while addressing some of their remaining limitations, particularly regarding dataset freshness, reproducibility, documentation, and labelling.

The contribution of this research is twofold. The first contribution is the dataset itself: an up-to-date dataset covering modern attacks against recent CVEs from the period 2020–2026. The dataset is created by attackers performing attacks on vulnerable machines in a controlled environment and capturing raw packets, NetFlow data, and system logs from the same events. The second part and probably the most important one in the long run is the methodology. A well-documented, repeatable way of creating attack datasets, covering everything from the environment set up and monitoring to running the attacks, time synchronization, and labeling the data automatically.

The motivation for this work is to support the development of more robust cybersecurity systems by providing a dataset that actually reflects what's happening currently in the field and leaving behind a methodology that can be used to create up-to-date datasets in the future.

2. План на работа и очекувани резултати

Во овој дел се внесува план на работа во кој се опишани фазите на изработка на магистерскиот труд. (Пр. преглед на релевантните методи од областа, идентификување на проблеми и насоки и идеи за нивно решавање и сл.)

The work on the master thesis will be implemented in a number of phases. Starting from research, to planning, deployment and conducting the experiment at the end.

The initial research phase will focus on reviewing related work in the field of intrusion detection datasets. This includes analysing existing datasets, identifying their strengths and limitations, and extracting lessons relevant to the design of a new dataset generation methodology. A clear understanding of previous work will help identify methodological gaps, design requirements, and potential risks that should be addressed in the proposed approach.

The overall process should begin with planning the initial structure of the methodology. The methodology will be organised around several key aspects and then make decisions on the questions that arise in each one of them.

The structure should be:

- Scenario design: defining the infrastructure, the types and number of vulnerable machines, and the number of attackers participating in the experiment.
- Monitoring: selecting and configuring tools for raw packet capture, system-log collection, NetFlow generation, and Network Time Protocol synchronisation.
- Attacks execution: defining the types of attacks, the target vulnerabilities, the execution procedure, the attack distribution, and the documentation required for reproducibility.
- Automatic Labelling: designing an automated mechanism for labelling the recorded data based on structured attack logs and synchronised telemetry.

There is also a need to define the attackers VM's. The idea is for each attacker to receive its own virtual machine inside the controlled environment. Based on all the attacks and exploits that need to be performed, a VM image should be designed and created containing all necessary tools and utilities for conducting the required attacks.

The next similar phase is defining the vulnerable machines. Since a broader spectrum of vulnerabilities is planned, VM images of vulnerable machines need to be defined and created for use. For example, Metasploitable 2, Metasploitable 3, and VM images with different combinations of VulHub docker containers.

In the writing and testing of the attack logging tool phase, such must be developed for use by attackers. Clear documentation and instructions on when and how to use the tool must also be provided.

In addition to this, creating and providing instruction to the attackers on how, when and what attack to conduct is also necessary.

Then comes the deployment and configuration phase. All parts of the infrastructure must be deployed and configured in order to prepare the environment for the experiment.

The main phase where the actual experiment will happen is the conducting of the experiment and recording the traffic. The attackers will perform dedicated attacks on the vulnerable VMs in the controlled environment, and all the traffic and system logs will be recorded for future use.

After the experiment is completed, the automatic labeling phase follows, in which a tool for automatic labeling is designed and developed based on attack log data generated by the attackers' CLI attack logging tool. This tool should be fast and accurate in order to provide the best results.

After the tool is created and tested, automatic labeling is performed on the recorded data.

Following the completion of automatic labelling, a validation step will be performed on the resulting dataset to assess its quality and correctness. This includes verifying label accuracy by cross-referencing a sample of labelled records against the raw attack logs, checking for time-synchronization consistency across packet, flow, and log data, and inspecting the dataset for common issues such as class imbalance, missing values, and duplicate records. The goal is to ensure that the dataset is reliable and suitable for use in machine learning-based intrusion detection research before it is finalized and published.

The expected contributions of this master thesis are as follows. A set of attack scenarios based on recent vulnerabilities, defining a list of attacks identified by unique names and CVE identifiers. A controlled experimental environment for generating attack traffic, consisting of an isolated network infrastructure with individually assigned attacker virtual machines and a group of vulnerable target machines, on which the experiment is conducted. A logging mechanism for tracking attack execution in the form of a CLI tool used by attackers to track every action taken during the experiment. An automatic labelling process implemented with a tool that takes the structured data from the attackers' actions and uses it to label the raw collected data with precision. A labelled dataset containing packet, flow, and log data, combining raw network packet captures, derived network flow records, and system-level log entries, all time-synchronized and labelled. Finally, a documented methodology for reproducing the process, providing a complete, step-by-step explanation of every phase, from

environment setup and attack definition to traffic recording and automatic labelling, and with that, the process can be replicated and extended by other researchers.

3. Извадок од литература што се планира да се користи

Во овој дел се внесува листа на референтна литература која е планирано да се користи.

1. Engelen, G., Rimmer, V., & Joosen, W. (2021). Troubleshooting an Intrusion Detection Dataset: the CICIDS2017 Case Study. <https://ieeexplore.ieee.org/document/9474286>
2. Ferrag, M. A., Maglaras, L., Moschoyiannis, S., & Janicke, H. (2020). Deep Learning for Cyber Security Intrusion Detection: Approaches, Datasets, and Comparative Study. <https://www.sciencedirect.com/science/article/abs/pii/S2214212619305046>
3. Rosay, A., Cheval, E., Carlier, F., & Leroux, P. (2022). Network Intrusion Detection: A Comprehensive Analysis of CIC-IDS2017. In Proceedings of ICISSP 2022. <https://www.scitepress.org/Papers/2022/107740/pdf/index.html>
4. Jindal, R., & Anwar, A. (2021). Emerging Trends of Recently Published Datasets for Intrusion Detection Systems (IDS): A Survey. <https://arxiv.org/abs/2110.00773>
5. Almseidin, M., Al-Sawwa, J., & Alkasassbeh, M. (2022). Generating a Benchmark Cyber Multi-Step Attacks Dataset for Intrusion Detection. Journal of Intelligent & Fuzzy Systems. <https://journals.sagepub.com/doi/abs/10.3233/JIFS-213247>
6. Landauer, M., Skopik, F., Frank, M., Hotwagner, W., Wurzenberger, M., & Rauber, A. (2023). Maintainable Log Datasets for Evaluation of Intrusion Detection Systems. IEEE Transactions on Dependable and Secure Computing. <https://ieeexplore.ieee.org/document/9866880>
7. Pellegrino, G., et al. (2025). A Review on Intrusion Detection Datasets: Tools, Processes, and Features. Computer Networks. <https://www.sciencedirect.com/science/article/pii/S1389128625001458>
8. Goldschmidt, P., & Kučera, J. (2025). Network Intrusion Datasets: A Survey, Limitations, and Recommendations. <https://arxiv.org/abs/2502.06688>
9. Cantone, M., et al. (2024). Machine Learning in Network Intrusion Detection: A Cross-Dataset Generalization Study. IEEE Access. <https://doi.org/10.1109/ACCESS.2024.3472907>
10. Feretzakis, G., et al. (2025). Evaluating Machine Learning-Based Intrusion Detection Systems with Explainable AI: Enhancing Transparency and Interpretability. Frontiers in Computer Science. <https://doi.org/10.3389/fcomp.2025.1520741>

Студент

Јован Симоноски

(име, презиме и потпис)

Скопје, 15.05.2026